Exact Bayesian Inference for Spike-and-Slab Priors in Takagi–Sugeno–Kang Fuzzy Systems with Approximately Calibrated Model-Averaged Prediction Intervals

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Abstract

Calibrated prediction intervals are essential for decision-making under uncertainty, and Bayesian model selection is a principled route to them for rule-based regressions with linear consequents. However, replacing posterior sampling with an analytical approximation—BIC-based inclusion probabilities, hard thresholding, and a Laplace predictive covariance—can collapse accuracy. In the implementation we study, coverage collapses on ill-conditioned design matrices. We introduce exact inference (up to finite Monte Carlo error) for spike-and-slab priors over Takagi–Sugeno–Kang (TSK) consequents. A block-Gibbs sampler computes the per-rule inclusion full conditional in $O(n(d+1)^2)$ time via matrix-determinant and Woodbury identities; Bayesian model averaging (BMA) then propagates model uncertainty into the predictive variance. On two UCI benchmarks (three targets), this yields near-nominal 95% intervals (PICP = 0.93–0.95) where the analytical approximation we study undercovers (PICP = 0.00–0.18). Interval widths are comparable to the closed-form conjugate baseline's on the Energy targets. A corrected dense baseline—a membership-function fix raising R^2 from 0.48 to 0.94 on the Energy-Cooling target—anchors the comparison. The approach applies to any rule-based regression with linear consequents, and the code and corrected baselines are released for reproducible, uncertainty-aware inference.

Keywords: uncertainty quantification, prediction intervals, Bayesian inference, spike-and-slab prior, Bayesian model averaging, Takagi–Sugeno–Kang fuzzy system

1. Introduction

Predictive models in engineering increasingly need more than point estimates: designers require calibrated uncertainty to support decisions under limited data. Rule-based regressions with linear consequents—of which Takagi–Sugeno–Kang (TSK) fuzzy systems [1] are a canonical instance—are attractive here: they combine nonlinear approximation with a compact, interpretable architecture. Their linear consequents also make Bayesian treatment especially natural, yielding posterior distributions, prediction intervals, and principled model selection. Building-energy simulation exemplifies the data regime that motivates this work: each simulation run is computationally expensive, yielding training sets of only hundreds of samples [2].

The Bayesian TSK literature has been hampered by two problems. The first is methodological. Spike-and-slab priors [3, 4] place positive probability on exact zero coefficients. A natural alternative to Markov chain Monte Carlo (MCMC) is analytical model selection—BIC-based inclusion probabilities followed by hard thresholding and a Laplace approximation to the predictive covariance. This approximation fails in two ways. Hard thresholding at posterior inclusion probability 0.5 discards rules that carry real (if partial) signal and collapses accuracy. The pipeline as commonly implemented—a ridge-regularized fit of the ill-conditioned design matrix combined with a Laplace predictive covariance on the pruned model—produces prediction intervals that cover only 0–18% of test points at a nominal 95% level. A faithful version of thresholding plus the Laplace covariance, fit by ordinary least squares, covers near-nominally but remains inaccurate and inefficient (Supplementary, Section 5). The approximation thus sacrifices accuracy, calibration, or both in exchange for a sparsity that, on the relevant benchmark, is not warranted.

The second problem is a *reproducibility* one: a subtle implementation error in the construction of fuzzy membership functions propagates through every downstream baseline. Specifically, an implementation pattern that computes the spreads of Gaussian membership functions from the training data during model fitting, but then *recomputes* them from query data (with incorrect cluster labels) at prediction time, silently depresses dense TSK baselines. The fuzzy partition at prediction time no longer matches the one learned during training. We reproduce this pattern, quantify its effect, and show that correcting the bug raises the dense TSK least-squares baseline on the Energy Efficiency cooling target from $R^2 = 0.48$ to 0.94 (Section 5.1), providing a concrete corrected benchmark against which the methodological contribution below is validated.

We replace the analytical approximation with exact MCMC inference. We implement a block-Gibbs sampler for the spike-and-slab prior and predict by Bayesian model averaging (BMA), in which the predictive distribution is a mixture over posterior draws rather than a point estimate from a single selected model. Averaging over rule configurations retains partial information from uncertain rules and carries model uncertainty into the predictive variance, which avoids both the accuracy collapse and the interval collapse. We show that this recovers near-nominal prediction intervals, PICP = 0.93–0.95 at the nominal 95% level, where the analytical approximation it replaces undercovers severely as implemented (PICP = 0.00–0.18). Interval widths are comparable to the closed-form conjugate baseline's on the Energy targets. To our knowledge, no prior work provides exact posterior sampling for spike-and-slab priors over rule-model consequents together with model-averaged prediction intervals.

Our contributions are threefold:

1. **Exact spike-and-slab inference for linear-consequent rule models.** A block-Gibbs sampler whose per-rule inclusion full conditional is computed in $O(n(d + 1)^2)$ time, with Bayesian model averaging (BMA) propagating model uncertainty into the predictive variance.
2. **Calibrated model-averaged prediction intervals and a diagnostic of analytical approximations.** Gibbs sampling with BMA recovers near-nominal 95% intervals (PICP = 0.93–0.95), in contrast to our implementation of the BIC-plus-Laplace pipeline, which undercovers (PICP = 0.00–0.18); an ablation isolates the failure, showing that hard thresholding collapses accuracy while the as-implemented pipeline's ridge-regularized fit of the ill-conditioned design matrix collapses coverage.
3. **When Sparse Selection Is Warranted.** We delineate when sparse selection is warranted: on low-dimensional data ($d = 8$) neither rule-level nor coefficient-level sparsity improves on the dense baseline. A real high-dimensional check ($d = 81$, Supplementary Table 7) reproduces the pattern rather than overturning it. The boundary is therefore not a low-dimensional artifact, and sparsity claims in this setting must be demonstrated rather than assumed.

Section 2 reviews related work. Section 3 presents the TSK model, the membership bug, and the Gibbs-based Bayesian estimators. Section 4 describes the experimental setup. Section 5 reports the corrected baseline, the main comparison, calibration, and the sparsity boundary. Section 6 discusses implications and limitations, and Section 7 concludes.

2. Related Work

2.1. Bayesian TSK Fuzzy Systems

Gu and colleagues introduced Bayesian TSK classifiers with conjugate Gaussian–Wishart priors [5], later extended to joint structure identification [6], imbalanced classification [7], and probabilistic Gaussian centralization [8]. A zero-order Bayesian TSK variant is studied in [9], prediction intervals for adaptive neuro-fuzzy inference systems have been constructed through interval-network training in [10], and evolving fuzzy systems maintain interval predictions in nonstationary environments [11]. None of these targets sparse high-order consequents through posterior model averaging. These frameworks provide full posterior uncertainty estimates (posterior variances and prediction intervals) but employ non-sparsity-inducing priors, so every rule remains active regardless of relevance [12]. Our work builds directly on this line: the dense conjugate posterior (Section 3.3) is the natural non-sparsity-inducing reference against which sparse Bayesian variants should be judged, and we use it to validate our MCMC implementation.

2.2. Sparse TSK and Bayesian Variable Selection

On the frequentist side, Gong et al. [13] combine LASSO with TSK consequents for embedded feature selection, and Zhang et al. [14] use group-LASSO regularization for nonstationary TSK networks; convergence guarantees are studied in [15]. These methods yield point estimates without uncertainty quantification. Exact posterior computation for Bayesian model selection is generally intractable, and analytical surrogates built on information criteria or Laplace approximations have well-documented limitations [16, 17]. Recent work makes the error of Laplace approximations quantitative and finitely computable [18]. On the Bayesian side, spike-and-slab priors [3, 4, 19] have been applied to structurally sparse Bayesian neural networks via variational inference [20] and to high-dimensional sensor selection in engineering prognostics [21]. Denœux's ENNreg [22] provides a belief-function alternative to probabilistic fuzzy regression with prediction intervals but without rule selection. High-dimensional TSK systems target regimes with far more features: Bian et al. [23] introduce a multi-head TSK system for high-dimensional data, and Xue et al. [24] combine feature selection with rule extraction. Our low-dimensional benchmarks ($d = 8$) do not exercise this regime; we return to this contrast in the sparsity-boundary discussion (Section 5.4). Distribution-free alternatives such as *split* conformal prediction [25, 26] provide finite-sample coverage without posterior sampling. It requires a separate calibration set and does not decompose predictive variance into noise and model-selection components, which our BMA construction propagates explicitly. Our contribution is to implement exact (MCMC) spike-and-slab inference for TSK consequents with BMA-calibrated prediction intervals, which to our knowledge has not been developed for linear-consequent rule models, and to evaluate it against the dense baseline.

2.3. Reproducibility in Soft Computing

Reproducibility failures arising from silent implementation errors are a recognized concern in applied machine learning [27]. Within fuzzy systems, the practical consequences of an inconsistent fuzzy partition between training and inference have not, to our knowledge, been quantified. We quantify them on a reproduction of this pattern and release the corrected code and baselines publicly.

3. Methodology

3.1. TSK Fuzzy System

For input $\mathbf{x} \in \mathbb{R}^d$ and scalar target y, a TSK system with R rules is

$$y = \sum_{j=1}^{R} w_j(\mathbf{x})\,(\beta_{j0} + \beta_{j1}x_1 + \cdots + \beta_{jd}x_d) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \tag{1}$$

where $w_j(\mathbf{x}) = \prod_{i=1}^{d} \mu_{A_{ji}}(x_i) / \sum_{k=1}^{R} \prod_{i=1}^{d} \mu_{A_{ki}}(x_i)$ is the normalized firing strength of rule j, and A_{ji} are Gaussian antecedent fuzzy sets.

$$\mu_{A_{ji}}(x_i) = \exp\Big(-\tfrac{(x_i - c_{ji})^2}{2s_{ji}^2}\Big), \tag{2}$$

Centers c_{ji} are obtained from fuzzy c-means clustering [28], and spreads s_{ji} are set to the standard deviation of the training points assigned to each cluster, scaled by a factor s. Writing the weighted design matrix $\mathbf{\Phi} \in \mathbb{R}^{n \times R(d+1)}$ and stacking consequents into $\boldsymbol{\beta} \in \mathbb{R}^{R(d+1)}$, Eq. (1) becomes the linear model $\mathbf{y} = \mathbf{\Phi}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$.

3.2. The Membership-Spread Bug

The spreads s_{ji} in Eq. (2) define the fuzzy partition and must be identical at training and inference time. A defect we reproduce recomputes them at prediction: it calls the membership routine on query inputs while passing a placeholder label vector (e.g., all zeros), so the routine re-estimates spreads from the query data with degenerate cluster assignments. The resulting partition no longer matches the one used to fit $\boldsymbol{\beta}$. In our reproduction, this single defect depresses the dense TSK baseline from $R^2 = 0.94$ to 0.48 on the Energy-Cooling target (Section 5.1). The correct procedure stores the training-time centers and spreads and reuses them verbatim at inference, as in Algorithm 1.

Algorithm 1 Corrected TSK training and inference

Require: Training data $\{\mathbf{x}_i, y_i\}_{i=1}^n$, rule count R

1: $(\mathbf{c}, \mathbf{u}) \leftarrow \text{FCM}(\mathbf{X}, R)$ ▷ fuzzy c-means
2: $\mathbf{s} \leftarrow$ {per-cluster standard deviations from training points, scaled by s}
3: Store $(\mathbf{c}, \mathbf{s})$ ▷ *fix the partition at training time*
4: Build $\mathbf{\Phi}$ via Eqs. (1)–(2) using $(\mathbf{c}, \mathbf{s})$
5: Fit $\boldsymbol{\beta}$ (least squares, conjugate Bayes, or Gibbs; Sections 3.3–3.4)
6: **return** model with frozen $(\mathbf{c}, \mathbf{s}, \boldsymbol{\beta})$

3.3. Conjugate Bayesian Baseline

The non-sparse Bayesian TSK places a Gaussian prior $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \tau^2\mathbf{I})$ and an inverse-gamma prior $\sigma^2 \sim \text{InvGamma}(0.01, 0.01)$ on the noise variance. The posterior is available in closed form: with $\mathbf{P} = \mathbf{\Phi}^\top\mathbf{\Phi} + \tau^{-2}\mathbf{I}$,

$$\boldsymbol{\beta} \mid \mathcal{D}, \sigma^2 \sim \mathcal{N}(\mathbf{P}^{-1}\mathbf{\Phi}^\top\mathbf{y},\ \sigma^2\mathbf{P}^{-1}), \tag{3}$$

and the predictive variance at a new design row $\boldsymbol{\phi}^*$ is $\sigma^2 + \boldsymbol{\phi}^*(\sigma^2\mathbf{P}^{-1})\boldsymbol{\phi}^{*\top}$. This closed-form model serves both as a calibrated reference and as a correctness check for our sampler (a Gibbs sampler with all rules forced active must recover it). When the block-Gibbs sampler of Section 3.4 is run with all inclusion indicators fixed to $\gamma_j = 1$, the sampled predictive distribution recovers the conjugate closed form on held-out data. The predictive mean matches in R^2 to within 10^{-3} (measured 0.0010 at the default $R = 5$), and the predictive variance agrees to within ~6% (median; measured 5.6%). These results confirm the correctness of the $\boldsymbol{\beta}$ and σ^2 updates against the closed form (reproduction log in `results/raw/gibbs_verify_output.txt`). The inclusion-indicator full conditional of Eq. (5) is not exercised by this fixed-activation check. We validate it separately on a synthetic problem with known sparse ground truth, where the sampler's inclusion probabilities agree with exhaustive model averaging to within 10^{-3} (Supplementary, Section 9). We compare at the predictive level: the normal-equations matrix $\boldsymbol{\Phi}^\top\boldsymbol{\Phi}$ is numerically ill-conditioned (condition number $\sim 10^{18}$), which makes the raw posterior covariance numerically unstable for both the closed form and any MCMC estimate. The regularized estimation problem is nevertheless well posed. The dense fits use ridge-regularized normal equations (penalty 10^{-4} on $\boldsymbol{\Phi}^\top\boldsymbol{\Phi}$), which stabilizes them.

3.4. Spike-and-Slab Prior and Gibbs Sampling

We decompose $\boldsymbol{\beta}$ into R blocks $\boldsymbol{\beta}_j \in \mathbb{R}^{d+1}$ and place a spike-and-slab prior on each block:

$$\boldsymbol{\beta}_j \mid \gamma_j, \tau^2 \sim \begin{cases} \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}_{d+1}), & \gamma_j = 1 \ \text{(slab)}, \\ \delta_{\mathbf{0}}, & \gamma_j = 0 \ \text{(spike)}, \end{cases} \tag{4}$$

with $\gamma_j \sim \text{Bernoulli}(\pi)$. Rather than approximate the posterior over $\boldsymbol{\gamma}$ with a BIC criterion, we sample it exactly with a block-Gibbs sampler. Here "exact" means that we target the posterior by sampling rather than replacing it with a BIC-plus-Laplace analytical surrogate; as with any MCMC procedure the estimates carry finite Monte Carlo error, which we quantify with the convergence diagnostics of Section 4. For block j, let $\mathbf{r}_{-j} = \mathbf{y} - \sum_{k \neq j} \boldsymbol{\Phi}_k \boldsymbol{\beta}_k$ be the residual excluding rule j, and let $\mathbf{A}_j = \boldsymbol{\Phi}_j^\top \boldsymbol{\Phi}_j$. The marginal likelihood of $\mathbf{r}_{-j}$ under $\gamma_j = 1$ (integrating out $\boldsymbol{\beta}_j$) is Gaussian with covariance $\sigma^2 \mathbf{I} + \tau^2 \boldsymbol{\Phi}_j \boldsymbol{\Phi}_j^\top$. Its determinant and quadratic form are computed in $O(n(d+1)^2)$ via the matrix-determinant and Woodbury identities. The full conditional for γ_j is

$$P(\gamma_j = 1 \mid \cdot) = \sigma\Big(\log \tfrac{\pi}{1-\pi} - \tfrac{1}{2} \log \det(\mathbf{I} + \tfrac{\tau^2}{\sigma^2} \mathbf{A}_j) + \tfrac{\tau^2}{2\sigma^4} \mathbf{z}_j^\top (\mathbf{I} + \tfrac{\tau^2}{\sigma^2} \mathbf{A}_j)^{-1} \mathbf{z}_j \Big), \tag{5}$$

where $\mathbf{z}_j = \boldsymbol{\Phi}_j^\top \mathbf{r}_{-j}$ and $\sigma(\cdot)$ is the logistic function. Conditional on $\gamma_j = 1$, we draw $\boldsymbol{\beta}_j \sim \mathcal{N}((\mathbf{A}_j/\sigma^2 + \tau^{-2}\mathbf{I})^{-1} \mathbf{z}_j/\sigma^2,\ (\mathbf{A}_j/\sigma^2 + \tau^{-2}\mathbf{I})^{-1})$; otherwise $\boldsymbol{\beta}_j = \mathbf{0}$. Finally σ^2 is drawn from its inverse-gamma full conditional. A coefficient-level variant applies the same prior independently to each scalar coefficient β_i (stochastic search variable selection, SSVS [4]), which is appropriate when feature-level rather than rule-level sparsity is of interest. For numerical stability on the ill-conditioned design matrix, the implementation adds a small diagonal jitter (10^{-10}) to the sampled precision matrices, initializes the coefficients from a ridge-regularized least-squares fit, and floors the noise variance at 10^{-4}.

3.5. Bayesian Model Averaging for Prediction

We predict by BMA rather than by conditioning on a single selected model [29]; recent reviews of BMA practice and its pitfalls are given in [30]. Given T posterior draws $\{\boldsymbol{\beta}^{(t)}, \sigma^{2(t)}\}_{t=1}^T$, the predictive mean is the mixture mean $\hat{y}^* = \frac{1}{T}\sum_t \boldsymbol{\phi}^* \boldsymbol{\beta}^{(t)}$, and the predictive variance follows from the law of total variance:

$$\mathrm{Var}(y^* \mid \mathbf{x}^*, \mathcal{D}) = \underbrace{\frac{1}{T}\sum_t \sigma^{2(t)}}_{\text{aleatoric}} + \underbrace{\mathrm{Var}_t(\boldsymbol{\phi}^* \boldsymbol{\beta}^{(t)})}_{\text{model uncertainty}}, \tag{6}$$

yielding the 95% interval $\hat{y}^* \pm 1.96\sqrt{\mathrm{Var}(y^* \mid \mathbf{x}^*, \mathcal{D})}$. Over posterior draws the second term spans both between-model (rule-configuration) variation and within-model parameter variation. It is precisely the model-selection uncertainty that a hard-thresholded approximation discards, and retaining it is what restores calibration.

4. Experimental Setup

We evaluate on two UCI regression benchmarks (three targets): the Energy Efficiency dataset [2] (768 simulated building configurations, 8 geometric features, with heating and cooling loads as two targets) and the Concrete Compressive Strength dataset [31] (1030 laboratory samples, 8 mix-design features). Features are standardized to zero mean and unit variance; all TSK methods use $R = 5$ rules from fuzzy c-means ($m = 2$) with Gaussian membership spreads equal to the per-cluster standard deviation of training points, scaled by $s = 1.5$. A rule-count sweep ($R = 3$–10, Supplementary) shows dense accuracy is nearly unchanged from $R = 4$ upward on the Energy targets and improves only marginally beyond $R \approx 7$ on Concrete, supporting the default $R = 5$.

Four TSK methods are compared in the main experiments: **TSK-LS** (dense least squares), **Bayesian-TSK** (dense conjugate posterior, Section 3.3), **TSK-SpikeSlab-BIC** (the BIC-plus-Laplace analytical approximation with hard thresholding; labeled `SpikeSlab-Fast` in the released code), and **TSK-SpikeSlab-Gibbs** (rule-level Gibbs plus BMA, Sections 3.4–3.5; labeled `SpikeSlab-Gibbs`). A coefficient-level variant, **TSK-SSVS** (labeled `SSVS-Gibbs`), is used in the sparsity-boundary analysis (Section 5.4). Random forest, support vector regression, and Gaussian-process regression serve as non-interpretable machine-learning references; the Gaussian process provides closed-form predictive intervals and is therefore the key probabilistic baseline for the calibration comparison. Reference implementations use the scikit-learn configurations shipped in the released code. A random forest with 300 trees and maximum depth 10, support vector regression with an RBF kernel ($C = 1$, $\gamma =$ `scale`), and a Gaussian process with an RBF-plus-white-noise kernel, hyperparameters optimized by marginal likelihood (`normalize_y=True`). Fuzzy-specific interval-construction methods such as ENNreg [22] and interval-ANFIS [10] build prediction intervals through evidential combination or interval-network training rather than posterior model selection. They are therefore not direct baselines for the model-selection question addressed here, and the Gaussian process—whose closed-form intervals come from a standard library implementation—serves as the probabilistic reference. The rule-level Gibbs sampler uses 1,000 burn-in and 2,000 retained draws; the coefficient-level SSVS uses 800 and 800. For the rule-level sampler, convergence is verified by running three chains with independent seeds on representative splits. The Gelman–Rubin statistic satisfies $\hat{R} \leq 1.001$ for the consequents and the noise variance, and the effective sample size exceeds 1,700 for the noise variance (Supplementary Table 6). The inclusion-indicator chains saturate near full activation on these benchmarks (Section 5.4), so their effective sample size equals the chain length by construction rather than from active exploration. Formal convergence verification is claimed only for the rule-level sampler, not for the coefficient-level SSVS runs. All results are averaged over 30 independent random

80/20 train–test splits with a fixed seed (SEED=42); reported intervals are 95% Gaussian intervals. All spike-and-slab priors use inclusion probability $\pi = 0.5$; the rule-level Gibbs and conjugate baselines use slab variance $\tau^2 = 10^3$, and the coefficient-level SSVS uses $\tau^2 = 1.0$. The $\pi = 0.5$ prior is dominated by the data on these benchmarks (posterior inclusion probabilities saturate near one), and the rule-level slab $\tau^2 = 10^3$ is deliberately diffuse relative to the standardized-feature scale. The sensitivity of the coefficient-level variant to τ^2 is characterized in Fig. 5a. Metrics are root-mean-square error (RMSE), R^2, prediction-interval coverage probability (PICP), and mean prediction-interval width (MPIW).

5. Results

5.1. A Corrected Dense Baseline

Fig. 1 contrasts the R^2 values reproduced under the buggy membership implementation with those obtained after correcting it. The improvement is consistent: TSK-LS rises from 0.48 to 0.94 (Energy-Cooling), 0.78 to 0.97 (Energy-Heating), and −2.86 to 0.74 (Concrete). The conjugate Bayesian-TSK rises from 0.78 to 0.94, 0.84 to 0.97, and −0.56 to 0.74. This confirms that the bug, not the model class, was responsible for the apparently mediocre baselines, and it establishes the corrected numbers as the reference for the comparisons below.

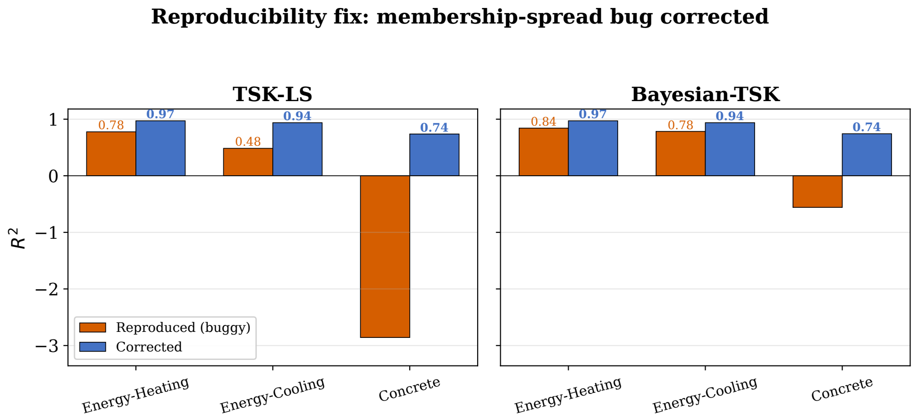

Figure 1: Corrected dense baselines. Reproduced R^2 under the buggy membership implementation (orange) versus corrected R^2 (blue, training-time membership spreads reused at inference) for the dense TSK-LS and conjugate Bayesian-TSK baselines. Values are means over 30 splits on the Energy (Heating and Cooling) and Concrete targets; correcting the membership-spread bug raises the dense TSK-LS R^2 from 0.48 to 0.94 on Energy-Cooling.

5.2. Main Comparison

Table 1 and Fig. 2 report the primary results (full mean-absolute-error results in Supplementary Table 1). After correcting the bug, the dense baselines are strong: TSK-LS achieves R^2 = 0.94–0.97 on the Energy targets and 0.74 on Concrete, and the conjugate Bayesian-TSK differs from it by at most 0.01 in R^2. The proposed Gibbs sampler recovers this accuracy: TSK-SpikeSlab-Gibbs attains R^2 = 0.97/0.94/0.74, matching the dense baseline to within 0.01 in R^2 on every target. This is the expected behavior of a correct sampler and a direct validation of our MCMC implementation against the closed form. A paired Wilcoxon signed-rank test over the 30 splits

[32] finds no significant difference between TSK-SpikeSlab-Gibbs and TSK-LS on the Energy targets (Heating $p = 0.11$, Cooling $p = 0.07$). It also finds no significant difference between Bayesian-TSK and TSK-LS (Heating $p = 0.32$, Cooling $p = 0.87$). The test does not establish equivalence, and the split-level results are not independent because the splits reuse the same underlying rows. On Concrete the difference is nominally significant ($p < 0.01$) but tiny in magnitude, with the samplers slightly above the dense baseline (mean $\Delta R^2 = 0.006$ for Gibbs, 0.004 for Bayesian-TSK).

By contrast, the BIC-based analytical approximation TSK-SpikeSlab-BIC collapses to negative R^2 (−1.8 to −6.0), significantly worse than every other method on all three targets (all $p < 0.001$). It predicts worse than a constant mean because hard thresholding discards rules that the correctly specified model needs.

The Gaussian-process reference [33] attains the accuracy of the non-interpretable Random Forest (GP $R^2 = 0.998/0.98/0.89$). Its closed-form 95% intervals slightly under-cover (PICP = 0.93–0.94), whereas the exact Bayesian TSK methods reach near-nominal coverage (PICP = 0.93–0.95) at the cost of wider intervals. On the Heating and Concrete targets the TSK intervals sit closer to the nominal level than the GP's, while on Cooling the two are nearly tied (TSK 0.93 vs GP 0.94).

Table 1: Main comparison (mean over 30 splits; ± std shown for RMSE; R^2 and PICP are means). PICP is the 95% interval coverage. Differences among TSK-LS, Bayesian-TSK, and TSK-SpikeSlab-Gibbs are at most 0.01 in R^2; paired Wilcoxon tests over the 30 splits are reported in the text.

	Energy-Heating			Energy-Cooling			Concrete		
Method	RMSE	R^2	PICP	RMSE	R^2	PICP	RMSE	R^2	PICP
TSK-LS	1.69±0.19	0.97	–	2.36±0.17	0.94	–	8.52±0.55	0.74	–
Bayesian-TSK	1.69±0.19	0.97	0.95	2.36±0.17	0.94	0.93	8.46±0.43	0.74	0.95
TSK-SpikeSlab-BIC	16.4±3.4	−1.8	0.00	18.8±3.0	−3.1	0.00	44.0±4.0	−6.0	0.18
TSK-SpikeSlab-Gibbs	1.70±0.19	0.97	0.95	2.37±0.17	0.94	0.93	8.43±0.43	0.74	0.95
Random Forest	0.49±0.06	0.998	–	1.67±0.13	0.97	–	5.17±0.40	0.90	–
SVR	2.73±0.21	0.93	–	3.09±0.22	0.89	–	9.97±0.43	0.64	–
Gaussian Process	0.47±0.05	0.998	0.93	1.35±0.11	0.98	0.94	5.53±0.46	0.89	0.94

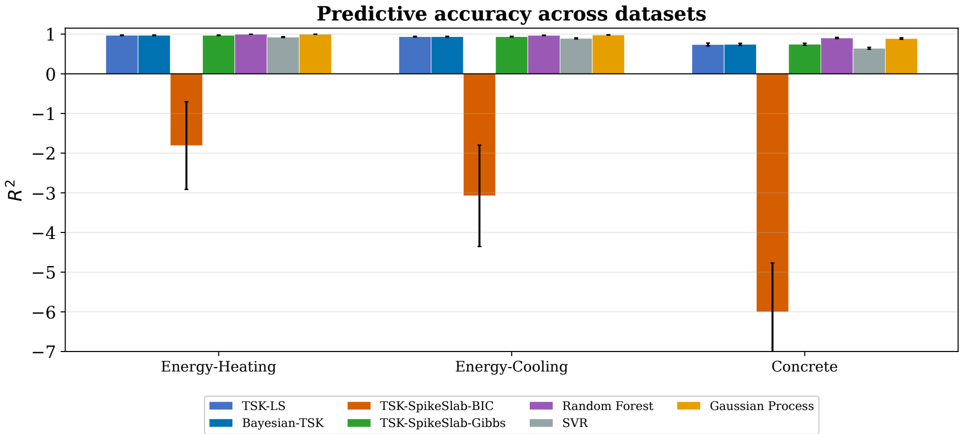

Figure 2: Predictive accuracy (R^2) across datasets. The Gibbs/BMA sampler (TSK-SpikeSlab-Gibbs) recovers the dense-baseline accuracy on all three targets (Energy-Heating, Energy-Cooling, Concrete; $R^2 = 0.97/0.94/0.74$), while the BIC-based analytical approximation (TSK-SpikeSlab-BIC) collapses to negative R^2.

5.3. Calibration

Fig. 3 reports interval calibration, assessed with the coverage and width diagnostics standard in the probabilistic-forecast literature [34]. Pooled over the 30 splits, both the conjugate Bayesian-TSK and the Gibbs sampler attain mean PICP of 0.95 on the Energy-Heating and Concrete targets and 0.935 on Energy-Cooling (displayed as 0.93 in Table 1). The 95% binomial (Wilson) intervals are $[0.945, 0.958]$, $[0.944, 0.955]$, and $[0.927, 0.941]$, respectively. The Gibbs values agree with the conjugate values to within 0.004. Coverage is therefore within one binomial standard error of the nominal 95% on Heating and Concrete, and about 1.5 percentage points below it on Cooling. The deviation lies within one per-split standard deviation (~ 0.027 on Cooling) yet is statistically detectable once the splits are pooled. We therefore describe the intervals as near-nominal rather than perfectly calibrated. Widths are compact (MPIW $\approx$ 6.7–32.9; Supplementary Table 8). The analytical TSK-SpikeSlab-BIC baseline undercovers severely as implemented: PICP = 0.00 on both Energy targets and 0.18 on Concrete. All but a small fraction of test points on the Energy targets, and the large majority on Concrete, fall outside the nominal 95% intervals. The ablation (Supplementary, Section 5) isolates the mechanism: hard thresholding alone collapses accuracy ($R^2 = 0.60$) while retaining near-nominal coverage with inflated width (MPIW 26.1). The coverage collapse thus arises from the as-implemented pipeline's ridge-regularized fit of the ill-conditioned design matrix rather than from the Laplace covariance per se. The Gibbs/BMA prediction of Eq. (6) restores accuracy and efficiency, propagating model uncertainty through the mixture variance. The value of averaging over configurations is most visible where the closed-form conjugate baseline degrades. On the high-dimensional check (Superconductivity, $d = 81$, Section 5.4, Supplementary Table 7), the Gibbs/BMA estimator maintains near-nominal calibration (PICP 0.940) where the dense conjugate baseline slightly undercovers (PICP 0.921). Fig. 4 confirms this calibration across a range of nominal levels and shows the 95% intervals of the two Bayesian methods against the held-out truth on a representative split.

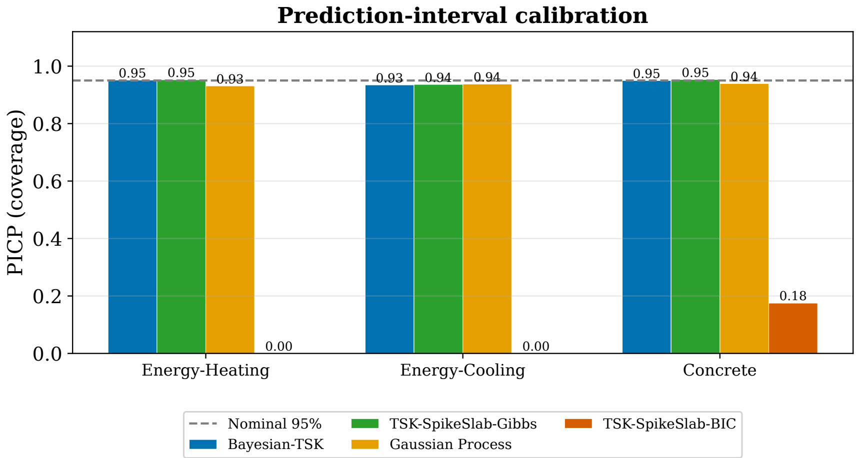

Figure 3: Prediction-interval coverage (PICP) with the nominal 95% reference (dashed). Gibbs/BMA and the conjugate Bayesian model attain near-nominal coverage; the BIC-plus-Laplace approximation undercovers as implemented.

5.4. The Sparsity Boundary

Having established that Gibbs/BMA is correct and calibrated, we characterize when the *sparsity* component of the spike-and-slab prior helps. Two results stand out (Fig. 5).

First, rule-level sparsity rarely activates: on all three targets the posterior inclusion probabilities of the $R = 5$ rules are close to 1.0 (with a single rule occasionally pruned on the Energy targets), because with a correct partition nearly every rule contributes. This is expected in the low-dimensional regime ($d = 8$), where the dense estimation problem is well posed. The rule-count sweep (Supplementary) confirms that even at $R = 10$ the inclusion probabilities remain near 1.0, so low dimensionality, not the rule count, keeps rule-level sparsity inactive.

Second, coefficient-level sparsity (SSVS) is a trade-off, not an accuracy gain. Varying the slab variance τ^2 over three orders of magnitude (Fig. 5a) shows that a tight slab ($\tau^2 = 0.1$) over-shrinks and collapses accuracy to $R^2 = -5.4$. A diffuse slab ($\tau^2 = 100$), in contrast, merely recovers the dense baseline ($R^2 = 0.92$ versus 0.94) while pruning nothing. No τ^2 setting improves on the dense baseline. In the irrelevant-feature regime (Fig. 5b), appending up to 30 standard-normal noise features leaves the dense baseline near its original level (ending at $R^2 = 0.88$, with a small non-monotone rise at four features that is within split-level noise). The sparse variants do not improve upon it. Correct fuzzy c-means already provides robustness to irrelevant features, so sparsity does not improve on the dense baseline in this low-dimensional regime.

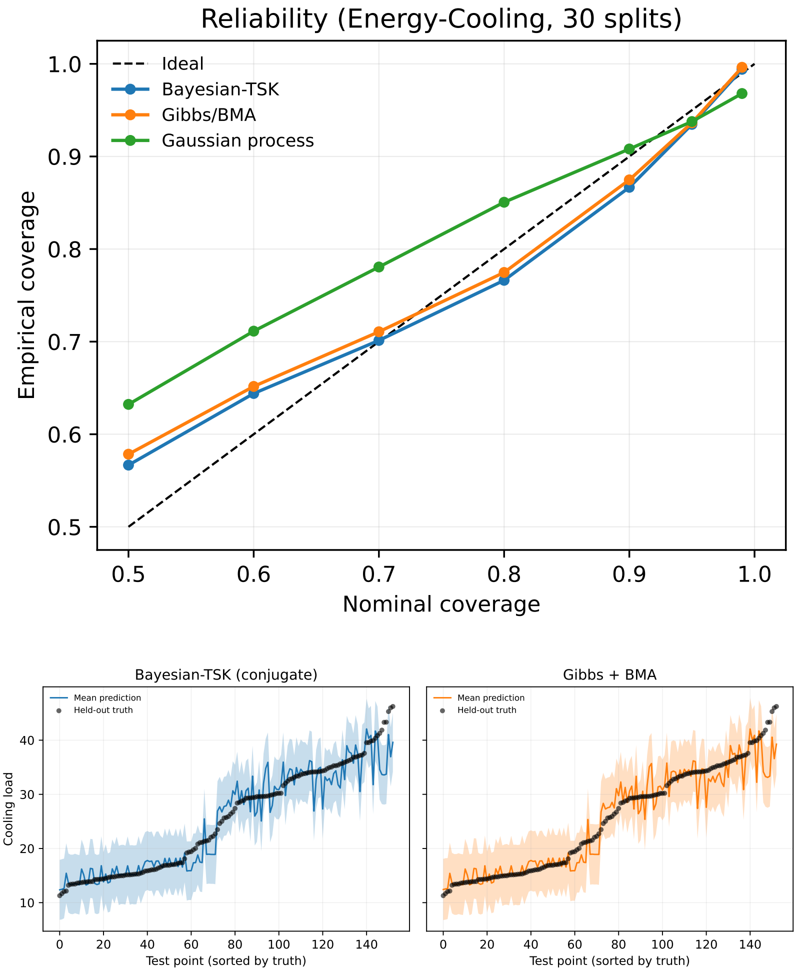

Figure 4: (a) Empirical coverage versus nominal coverage for the conjugate Bayesian-TSK, the Gibbs/BMA sampler, and the Gaussian-process reference on Energy-Cooling (30 splits). Points on the diagonal would indicate perfectly calibrated intervals; the Bayesian methods over-cover at the lowest nominal levels (empirical coverage 0.57 at nominal 0.5), under-cover slightly in the range 0.8–0.95 (empirical 0.94 at nominal 0.95), and return to the diagonal at nominal 0.99; the Gaussian process falls below the diagonal at nominal 0.99. (b) 95% prediction intervals (shaded) and mean prediction (line) versus held-out truth (points) on one representative Energy-Cooling split, for the conjugate Bayesian-TSK (left) and Gibbs+BMA (right). The intervals track the sorted truth closely and cover the large majority of test points at compact widths.

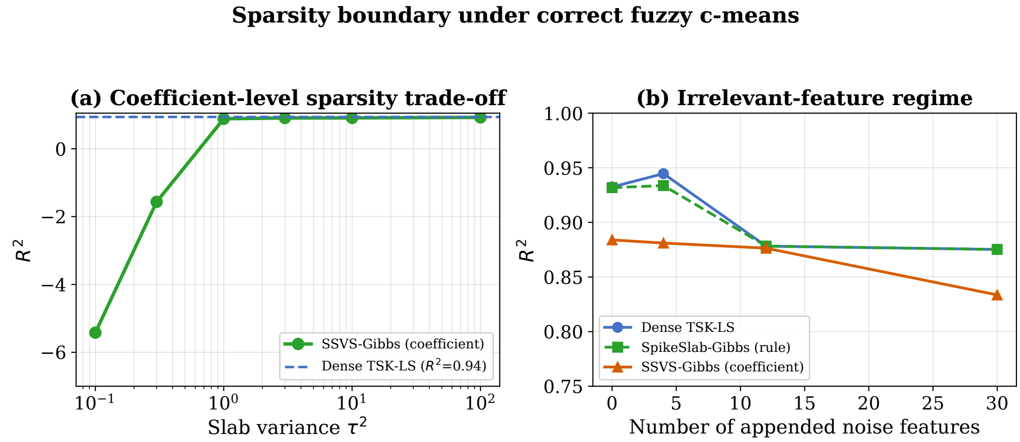

Figure 5: The sparsity boundary on Energy-Cooling ($R = 5$). (a) Coefficient-level SSVS R^2 versus slab variance τ^2 (10 splits): a tight slab over-shrinks and a diffuse slab only recovers the dense baseline (dashed; 30-split TSK-LS reference, $R^2 = 0.94$), never improving on it. (b) Irrelevant-feature regime (10 splits; dense baseline $R^2 = 0.93$ at zero noise features): the dense TSK-LS declines only modestly, from $R^2 = 0.93$ to 0.88, as noise features are appended. A small non-monotone rise at four features is within split-level noise. Neither rule-level nor coefficient-level sparsity improves on it (details in Supplementary Tables 2–3).

6. Discussion

6.1. Why the Analytical Approximation Fails

The failure of the BIC-plus-Laplace approximation has a specific origin. BIC approximates the Bayes factor [35] by replacing the integral over the slab with a point mass at the maximum-likelihood estimate. This is only reliable when the posterior over γ_j is concentrated at 0 or 1. On the Energy benchmark the inclusion probabilities are intermediate, so the hard threshold at 0.5 discards rules that carry real (if partial) signal. The Laplace predictive covariance then conditions on a single pruned model. With an ordinary-least-squares fit it alone yields wide intervals with near-nominal coverage (MPIW 26.1, PICP 0.93). The coverage collapse instead arises in the as-implemented pipeline, which fits the selected model with ridge regularization on the ill-conditioned design matrix. Gibbs sampling with BMA avoids both steps: it averages over all rule configurations rather than selecting one. An ablation isolates which step is decisive (Supplementary, Section 5): replacing hard thresholding with BMA over all 2^R rule subsets, using the same BIC-based model weights, restores dense-level accuracy and near-nominal calibration ($R^2 = 0.95$, PICP 0.92 on Energy-Cooling). Averaging rather than the BIC approximation is the decisive change for accuracy. Feeding the Gibbs sampler's inclusion probabilities through hard thresholding and the Laplace covariance performs equally well ($R^2 = 0.94$, PICP 0.94), confirming that the Gibbs inclusion probabilities are reliable. The as-implemented BIC baseline collapses ($R^2 = -2.5$, PICP 0.00; 10-split ablation, Supplementary Section 5, cf. −3.1 at 30 splits in Table 1). The coverage collapse is traceable to its ridge-regularized fit of the ill-conditioned design matrix.

6.2. Implications for Sparse TSK Research

Our boundary characterization carries a constructive message. Sparsity-inducing priors are worth their cost only when the model is over-parameterized (high-dimensional inputs with truly irrelevant features) or when the data are scarce relative to the number of effective parameters. Our real high-dimensional check ($d = 81$, Supplementary Table 7) does not realize this regime, since its features are largely relevant and $n = 3{,}000$ is not scarce relative to the effective parameter count. In the low-dimensional building-energy regime, the dense TSK is already well posed and no sparsity is warranted. The rule-count sweep (Supplementary) sharpens this: even at $R = 10$, where the model has $R(d+1) = 90$ consequent coefficients, rule-level sparsity does not activate. Rule count alone, without high dimensionality, does not make sparse selection advantageous on these benchmarks. This is not a failure of the spike-and-slab framework; it is a statement about the data, and it pinpoints where the framework's cost is justified. Alternative shrinkage priors, such as the continuous spike-and-slab LASSO [36] and the horseshoe [37], offer smooth regularizers that may behave differently under the collinearity of the TSK weighted design matrix. Extending the exact inference developed here to these priors is natural future work. We therefore present the Gibbs/BMA estimator as a *correct and calibrated* Bayesian TSK, and report sparsity as a boundary condition rather than a primary contribution.

6.3. Limitations

Three limitations scope this work. First, the benchmarks are low-dimensional ($d = 8$), although an evaluation on a real high-dimensional benchmark (Superconductivity, $d = 81$, subsample of $n = 3{,}000$ rows, Supplementary Table 7) reproduces the same pattern. Rule-level sparsity matches but does not improve on the dense baseline, and coefficient-level sparsity over-shrinks. The boundary is therefore not a low-dimensional artifact; whether sparsity is beneficial under different antecedent structures or much larger samples remains open. Second, we fix the antecedent structure and apply sparsity only to consequents; joint antecedent–consequent selection [6] remains future work. Third, our evaluation uses simulated (Ecotect) building-energy data; validation against measured, submetered energy data would strengthen the application case.

7. Conclusion

We introduced exact Bayesian inference for spike-and-slab priors over TSK consequents. A block-Gibbs sampler combined with Bayesian model averaging recovers near-nominal 95% prediction intervals (PICP = 0.93–0.95) where our implementation of the BIC-plus-Laplace analytical pipeline undercovers (PICP = 0.00–0.18). In practical terms, a modeler who needs a 95% interval from a rule-based model should average over rule configurations rather than commit to a single selected model. The averaged intervals track the nominal level closely, whereas the thresholded BIC baseline undercovers. We validated the method against a corrected dense baseline, obtained by fixing a membership-function bug that had depressed TSK baselines ($R^2 = 0.48 \rightarrow 0.94$ on the Energy-Cooling target). We also characterized when sparsity helps, showing that on low-dimensional data neither rule-level nor coefficient-level sparsity improves on the dense baseline, delineating the regime in which sparse selection is warranted. The approach applies to any rule-based regression with linear consequents, and the corrected code and baselines are released to support reproducible, uncertainty-aware inference.

CRediT authorship contribution statement

Zhiren Xiao: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Acknowledgments

The author declares no specific funding for this research.

Data Availability

Datasets are publicly available from the UCI Machine Learning Repository: Energy Efficiency (dataset 242) [2] and Concrete Compressive Strength (dataset 165) [31]. Code is available at `https://github.com/Jackxiaozhiren/tsk-spikeslab` and archived with the persistent DOI 10.5281/zenodo.21929319, including the corrected TSK core, experiment drivers, a pinned dependency manifest, and the fixed seed stated in Section 4.

Declaration of Competing Interests

The author declares no competing interests.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) for code development assistance and language editing. All scientific content, methodology design, experimental results, and interpretation were performed and verified by the author. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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